

Jase Branch

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SUMMARY

Embedded Systems & Robotics Engineer with hands-on experience in real-time firmware, autonomous systems, and sensor fusion. Proven ability to architect, simulate, and deploy complex systems spanning STM32, ROS2, and C++.

EDUCATION

University of Texas at Arlington

Arlington, TX

Bachelor of Science in Computer Science **GPA: 3.41/4.0**

May 2024

- **AI/Vision:** Computer Vision, Neural Networks & Deep Learning, Artificial Intelligence
- **Systems:** Computer Architecture, Operating Systems, Information Security

TECHNICAL SKILLS

Languages: C++, C, Rust, Python, Bash

Robotics & Middleware: ROS2, Nav2, SLAM, Gazebo

Embedded Systems: STM32, ESP32, FreeRTOS, CAN, UART, SPI, I2C, MQTT

Graphics & Compute: OpenGL, GLSL, CUDA, Bullet Physics

Networking & Systems: Boost.Asio, UDP/TCP, Real-Time Systems, Sensor Fusion

Tools & CI/CD: Git, CMake, Clang, Docker, Docker Compose, GitHub Actions

PROJECTS

Embedded 4-DOF Robotic Arm (C, Python, STM32, Fusion 360)

[GitHub](#) | [Blog](#) | Spring 2026

- Designed and fabricated a physical 4-DOF robotic arm using Fusion 360 for CAD modeling and 3D printing.
- Developed real-time STM32 firmware in C with PWM servo control and a 10 ms velocity-limited control loop.
- Consolidated power distribution onto a custom PCB, reducing wiring complexity and improving servo stability.
- Built a Python host interface with a UART state machine for scripted autonomous pose and gripper control.

Autonomous Drone Simulator (C++, OpenGL, Ada, Boost.Asio)

[GitHub](#) | [Blog](#) | Summer 2025

- Built a real-time autonomous drone simulator using Bullet Physics, custom rendering, and networked flight control.
- Emulated five onboard sensors (IMU, GPS, etc.) with **10-100 Hz** asynchronous updates.
- Measured system throughput exceeding **260 messages/sec** across the sensor and networking pipeline.
- Developed a virtual flight controller in both Ada and Python communicating over UDP with the simulation.

Embedded Sensor Fusion Network (C++, Rust, FreeRTOS, STM32, ESP32)

[GitHub](#) | [Blog](#) | Fall 2025

- Architected a multi-node embedded system streaming real-time IMU data over CAN and Wi-Fi.
- Tuned sensor pipeline sustaining **80-90 messages/sec** with low jitter.
- Reduced packet loss from **12% to under 3%** using multi-chunk CAN packetization and queue restructuring.
- Delivered reusable CAN serialization libraries and a Rust Embassy receiver sustaining **150-170 messages/sec**.

ROS2 Autonomous Navigation Benchmark (ROS2, Nav2, Gazebo, SLAM)

[GitHub](#) | [Blog](#) | Winter 2025

- Built a reproducible benchmarking framework for autonomous navigation using ROS2 and Nav2.
- Automated **4 parallel** navigation trials using Docker Compose for consistent and repeatable testing.
- Collected and analyzed metrics including completion time, path length, and trajectory efficiency.

GPU Boid Simulation (C++, CUDA, OpenGL)

[GitHub](#) | [Blog](#) | Spring 2025

- Implemented a GPU-accelerated Boid simulation using CUDA with OpenGL instanced rendering.
- Scaled performance from **600 to 20,000 agents** at **60 FPS**, achieving a **35-40×** speedup over CPU execution.
- Optimized GPU memory layout and kernel execution for large-population flocking behavior.

EXPERIENCE

Team Member — UT Arlington & Bell (Textron) Flight

Aug. 2023 – Dec. 2023

GPS-Denied Autonomous Drone Competition

Arlington, TX

- Collaborated with a 6-person multidisciplinary team and Bell engineers on a GPS-denied autonomous UAV system.
- Reproduced, diagnosed, and debugged intermittent MQTT communication failures observed during flight testing.
- Validated subsystem interfaces to ensure reliable end-to-end data flow across onboard and ground systems.

Blender / Smoothie Bowl Specialist

Jan. 2026 – Present

Mamaka Bowls

Fort Worth, TX

- Demonstrated reliability and teamwork in a fast-paced customer environment.
- Communicated effectively with coworkers to ensure efficient daily operations.